

Heartbeat-Evoked Potentials Reveal a Sleep-Stage and Age Gradient in Cortical Interoception

Nir Cafri^{1,3}, Felix Benninger^{2,3}, Pablo Blinder^{1,2}

¹Tel Aviv University; ²Sagol School of Neuroscience; ³Rabin Medical Center, Israel. Correspondence: nircafri@mail.tau.ac.il

Abstract

The heartbeat-evoked potential (HEP) is an EEG signature of cortical processing of cardiac afferent input, but its clinical use requires separating disease-related variation from normal physiological modulation by vigilance state and age. We analysed R-peak-locked EEG from a 19-electrode 10–20 clinical polysomnography cohort of 12,039 patients across three sleep stages (light sleep, N3, REM). A representative single-patient recording establishes the signal's origin relative to the cardiac-field artifact. Cohort-wide topographic maps show that HEP amplitude is largest over central and occipital sites and grows from light sleep to REM. Diagnosis groups (Obstructive Sleep Apnea (n=7,427), Hypertension (n=5,308), Obesity (n=4,093), Diabetes (n=3,283), Anxiety (n=3,203), Depression (n=3,104)) differed from an undiagnosed reference cohort within each sleep stage taken separately. Within a single diagnostic group followed across all three stages (suspected epilepsy, 2,075 matched patients), paired sleep-stage contrasts were significant for all 3 tested stage pairs after correction, and an age median-split produced spatially organized, stage-dependent differences. Together these results establish sleep stage and age as structured, separable axes of HEP variability that must be held constant when testing diagnosis as a clinical biomarker.

Introduction

The HEP is a small EEG deflection time-locked to the electrocardiographic R peak, interpreted as a marker of cortical interoception. It sits beside a much larger cardiac-field artifact and is sensitive to arousal state, so clinical interpretation requires (i) direct visualization of the source signals, (ii) exclusion of the peri-R interval, and (iii) comparison within a fixed sleep stage. Here we present, in five figures, the signal-level basis of the HEP, its cohort-wide topography, its dependence on clinical diagnosis within matched sleep stages, its within-patient modulation across sleep stages, and its association with age.

Signal-level origin of the HEP

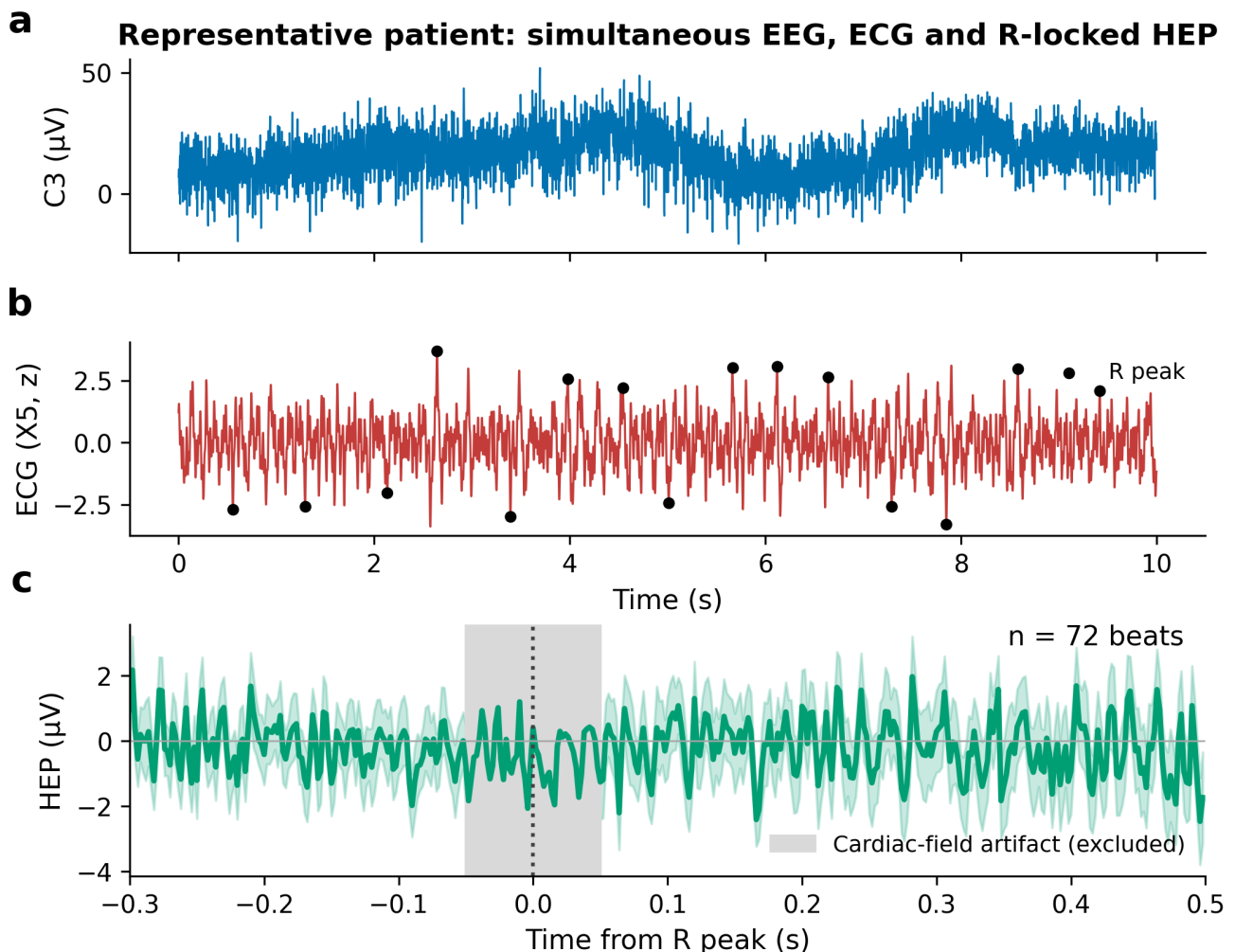
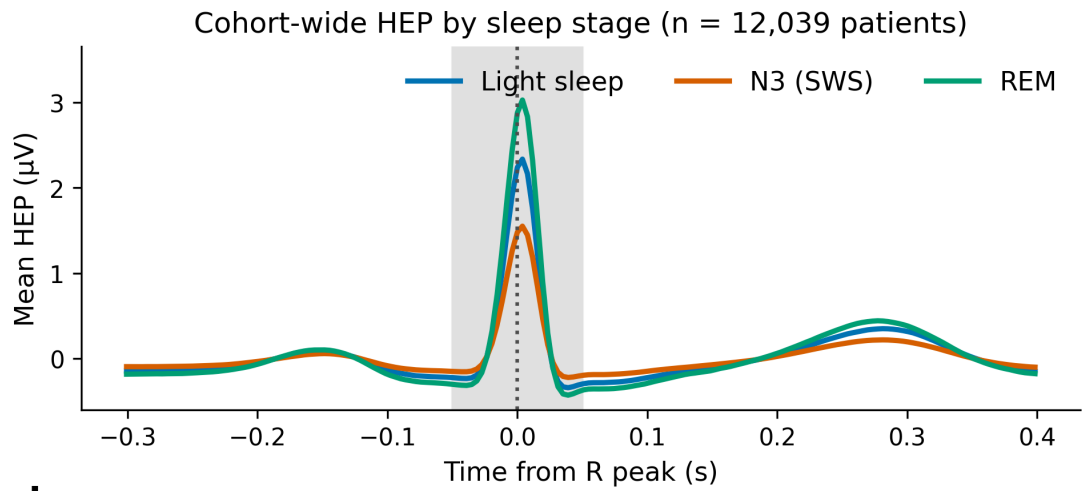


Figure 1 | Representative simultaneous EEG and cardiac activity from a single patient recording. R peaks (black markers) define epochs; the grey band marks the -50 to +50 ms cardiac-field-artifact interval excluded from inference. The lower trace is the patient-level mean HEP $\pm$ s.e.m. across beats.

Cohort-wide HEP amplitude and 19-electrode topography

a



b

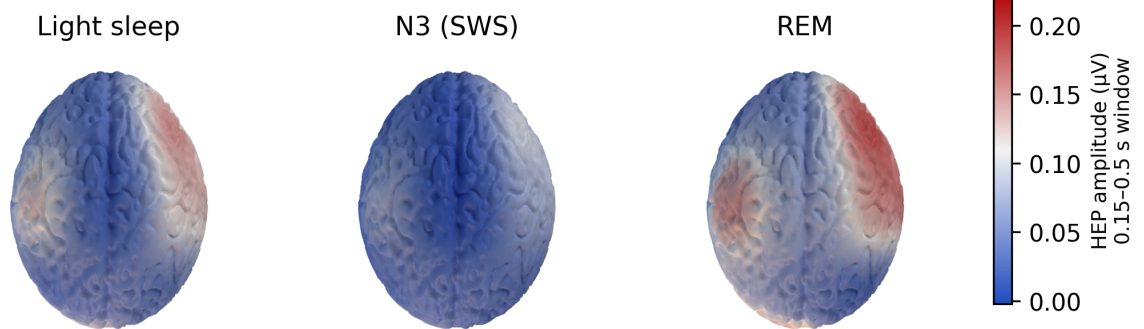


Figure 2 | (a) Grand-average HEP waveform by sleep stage across the full cohort. (b) 10–20 topographic maps of grand-mean HEP amplitude (0.15–0.5 s post-R window) at each of the 19 standard electrodes, separately for light sleep, N3 and REM.

Diagnosis effects within the same sleep stage

The most common diagnostic categories in the cohort were compared against an undiagnosed reference group separately within light sleep, N3 and REM, so that no comparison pools across vigilance states. Cluster-based permutation testing (peri-R artifact window excluded) identified significant amplitude windows for essentially every diagnostic group in every sleep stage, indicating that diagnosis-associated HEP differences are not an artifact of sleep-stage composition.

Grand-average HEP waveforms by diagnostic category, vs. undiagnosed reference cohort

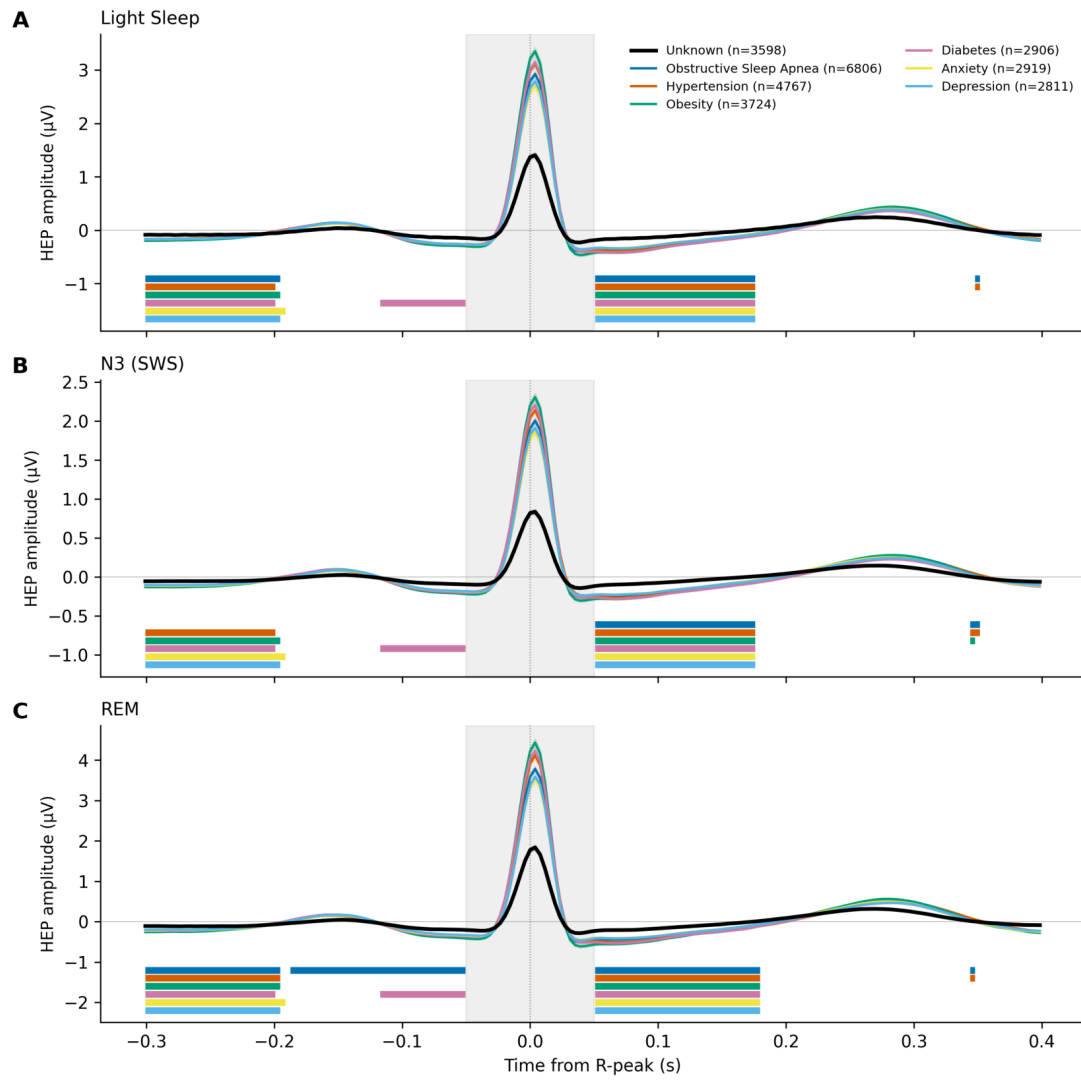
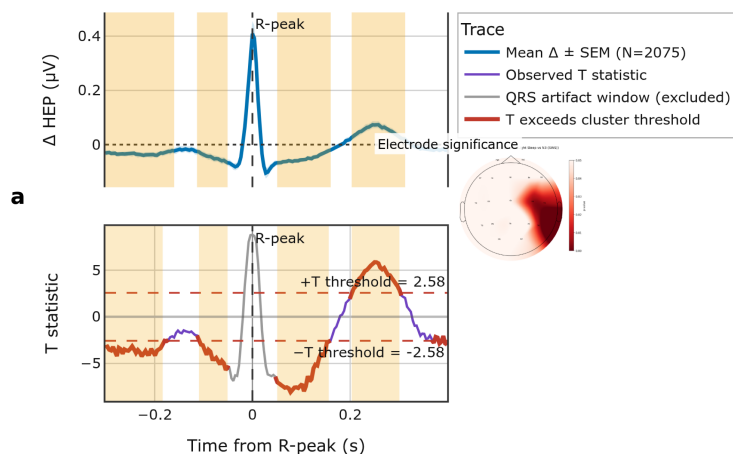


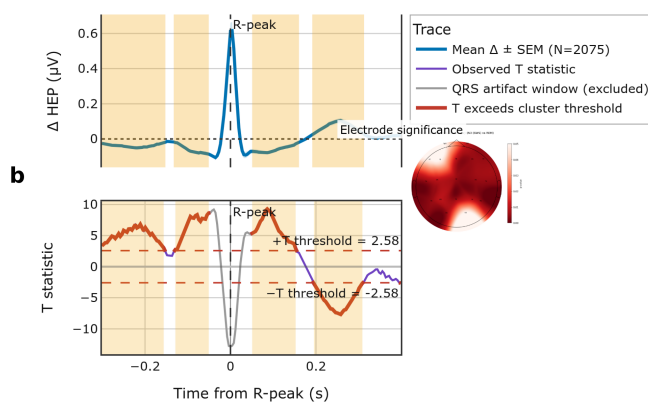
Figure 3 | Grand-average HEP waveforms by diagnostic category versus an undiagnosed reference cohort, shown separately within light sleep (a), N3 (b) and REM (c). Coloured horizontal bars mark time windows where a diagnostic group shows a significant cluster not also present in the reference cohort (cluster permutation test, $p < 0.05$); the grey band is the excluded cardiac-field-artifact window.

Within-diagnosis sleep-stage effects

Susp. Epilepsy: Light Sleep – N3 (SWS) (2075 matched patients)



Susp. Epilepsy: REM – N3 (SWS) (2075 matched patients) (auto-flipped: R-peak positiv)



Susp. Epilepsy: REM – Light Sleep (2075 matched patients) (auto-flipped: R-peak posit)

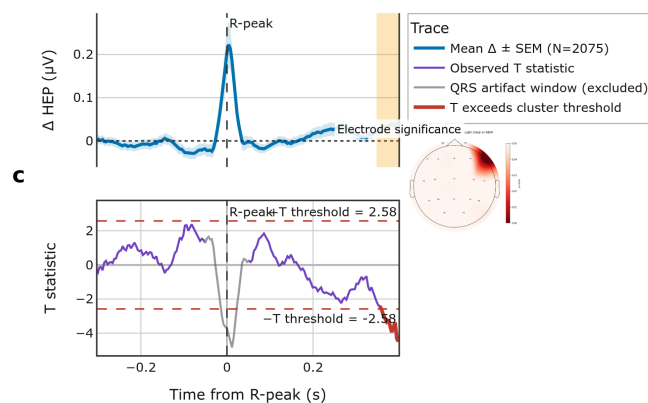
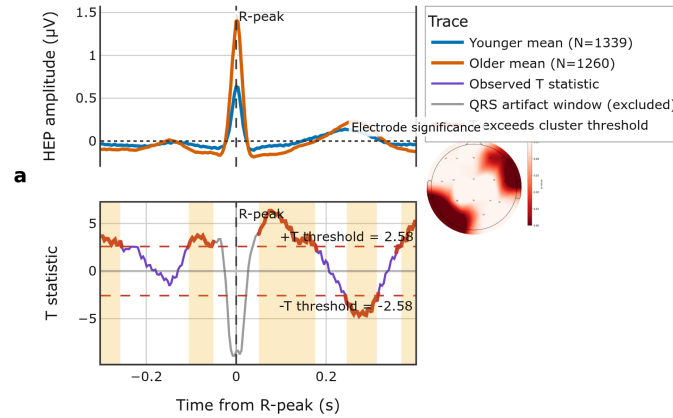


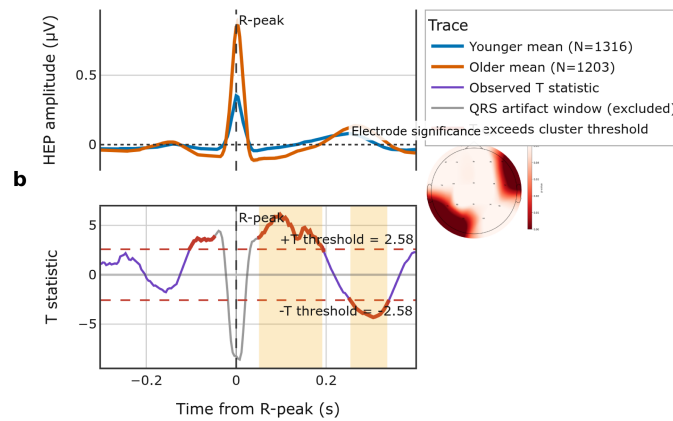
Figure 4 | Paired within-patient contrasts between sleep stages, restricted to a single diagnostic group (suspected epilepsy, 2,075 patients matched across all three stages) and shown only for the stage pairs that reached omnibus significance: (a) light sleep vs. N3, (b) REM vs. N3, (c) REM vs. light sleep. Each panel combines the mean difference waveform, its cluster t-statistic, and the electrode-wise significance topography; the peri-R grey interval is excluded from testing.

Age effects across sleep stages

Younger (N=1339) vs Older (N=1260): Light Sleep



Younger (N=1316) vs Older (N=1203): N3 (SWS)



Younger (N=1299) vs Older (N=1144): REM

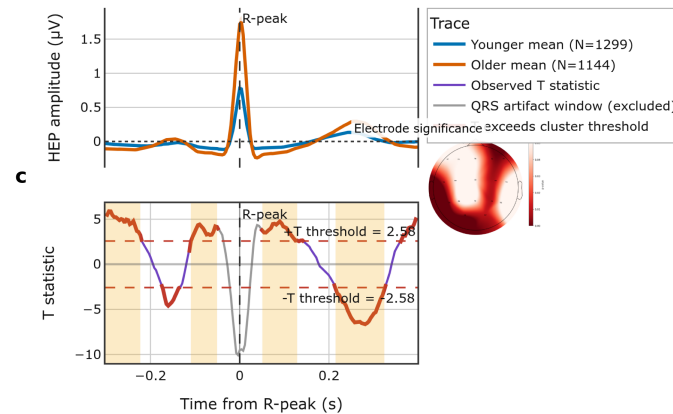


Figure 5 | Older-versus-younger median-split contrasts, shown only for sleep stages that reached the omnibus significance criterion: (a) light sleep, (b) N3, (c) REM. Waveform and 19-electrode maps show that ageing is a spatially organized covariate rather than a uniform amplitude offset.

Discussion

Three conclusions follow. First, a measurable post-R HEP is visible individually and cohort-wide once the artifact-dominated interval is excluded, peaking over central-occipital scalp. Second, diagnosis groups differ from an undiagnosed reference even within a single sleep stage. Third, sleep stage and age each produce large, topographically structured amplitude shifts that can confound unstratified disease comparisons. Future HEP biomarker studies should therefore be stage-matched, montage-matched and age-adjusted, with patient-level replication.

Methods summary

EEG was epoched around R peaks and baseline-corrected; the -50 to $+50$ ms interval was treated as cardiac-field artifact and excluded from inference. Diagnosis was tested within sleep stage against an undiagnosed reference using electrode-wise cluster-permutation tests, with heart rate and cardiac-field-artifact amplitude as covariates. This multi-hospital design parallels prior multicentre epilepsy imaging work.¹ Paired within-patient sleep-stage differences and age-split contrasts used cluster-mass permutation testing (200 permutations, cluster $\alpha = 0.01$), retaining only significant panels in Figures 4–5; Figure 2 used the same

frozen per-electrode amplitude table directly.

Acknowledgements

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References

1. Cafri N, Mirloo S, Zarhin D, Kamintsky L, Serlin Y, Alhadeed L, et al.; Benninger F. Imaging blood-brain barrier dysfunction in drug-resistant epilepsy: a multi-center feasibility study. *Epilepsia*. 2025;66(1):195-206.